

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Pipelining

Lecture No.- 6

By- Vishvadeep Gothi sir



Recap of Previous Lecture



doubt?

Topic

Data Hazard Classification

Topic

RAW, WAW, WAR

Topic

CPI Calculation



Topics to be Covered



Topic

CPI Calculation

Topic

Classical RISC Pipeline

#Q. Consider the following sequence of instructions, which is to be executed on a 5-stage pipeline: IF, ID, OF, EX and WB. Each stage takes one cycle for each instruction. Total Cycles required to execute this sequence

1. When pipeline uses operand forwarding $5 + 6 - 1 = 10$
2. When pipeline does not use operand forwarding $10 + 2 * (5 - 3) = 14$

ADD R2, R1, R0

MUL R4, R3, R2

SUB R6, R5, R4

ADD R6, R7, R8

MUL R7, R1, R2

SUB R1, R3, R4

R2 ← R1 + R0

R4 ← R3 * R2

R6 ← R5 - R4

R6 ← R7 + R8

R7 ← R1 * R2

R1 ← R3 - R4

#Q. Consider the following sequence of instructions, which is to be executed on a 5-stage pipeline: IF, ID, OF, EX and WB. Each stage takes one cycle for each instruction. Total Cycles required to execute this sequence

1. When pipeline uses operand forwarding $5 + 6 - 1 = 10 + 2 = 12$
2. When pipeline does not use operand forwarding $10 + 4 = 14$

ADD R2, R1, R0

R2 ← R1 + R0

MUL R4, R3, R2

R4 ← R3 * R2

LOAD R7, (1000)

R7 ← M[1000]

ADD R6, R7, R8

R6 ← R7 + R8

MUL R7, R1, R2

R7 ← R1 * R2

SUB R1, R3, R4

R1 ← R3 - R4

- #Q. Consider a non-pipelined processor operating at 2.5 GHz. It takes 5 clock cycles to complete an instruction. You are going to make a 5- stage pipeline out of this processor. Overheads associated with pipelining force you to operate the pipelined processor at 2 GHz. In a given program, assume that 30% are memory instructions, 60% are ALU instructions and the rest are branch instructions. 5% of the memory instructions cause stalls of 50 clock cycles each due to cache misses and 50% of the branch instructions cause stalls of 2 cycles each. Assume that there are no stalls associated with the execution of ALU instructions. For this program, the speedup achieved by the pipelined processor over the non-pipelined processor (round off to 2 decimal places) is 2.16.

$$S = \frac{t_n}{CPI_{avg} t_p}$$

$$= \frac{2 \text{ ns}}{1.85 * 0.5 \text{ ns}}$$

$$= 2.1621$$

pipeline:-

$$t_p = \frac{1}{2 \text{ GHz}} = 0.5 \text{ ns}$$



Non-pipeline:-

$$t_n = 5 * \frac{1}{2.5 \text{ GHz}}$$

$$= 2 \text{ ns}$$

$$CPI = 1 + (0.3 * 0.05 * 50) + (0.1 * 0.5 * 2)$$

$$= 1.85$$

[NAT]

PYQ GATE



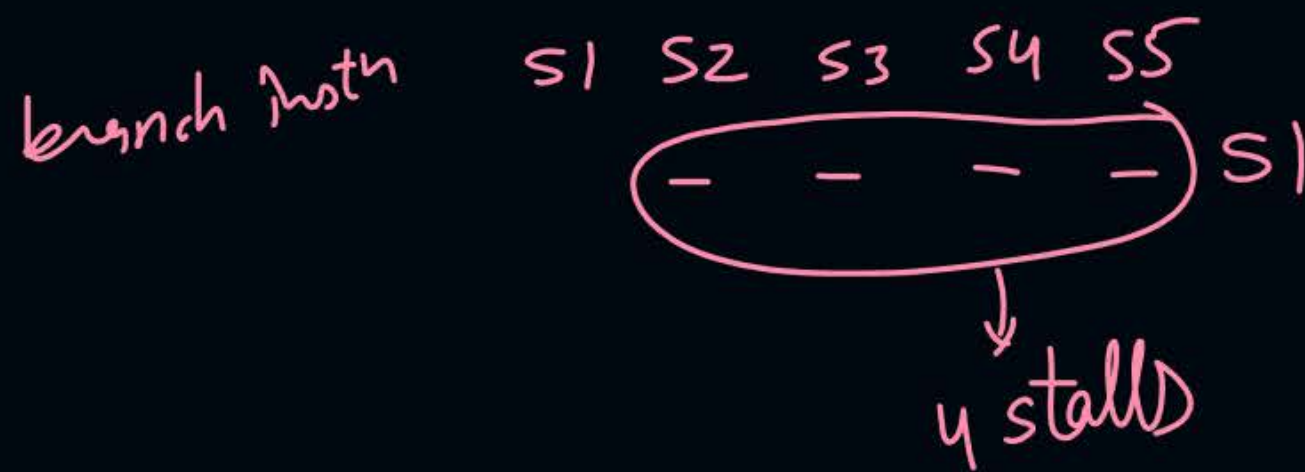
- #Q. An instruction pipeline has five stages where each stage takes $t_p = 2\text{ ns}$ (the number 2 is circled in red), and all instructions use all five stages. Branch instructions are not overlapped, i.e., the instruction after the branch is not fetched till the branch instruction is completed. Under ideal conditions.
1. Calculate the average instruction execution time assuming that 20% of all instruction executed are branch instructions. Ignore the fact that some branch instructions may be conditional. $\Rightarrow 3.6\text{ ns}$
 2. If a branch instruction is a conditional branch instruction, the branch need not be taken. If the branch is not taken, the following instructions can be overlapped. When 80% of all branch instructions are conditional branch instructions, and 50% of the conditional branch instructions are such that the branch is taken, calculate the average instruction execution time? 2.96

$$1. A.I.E.T. = CPI_{avg} * t_p$$

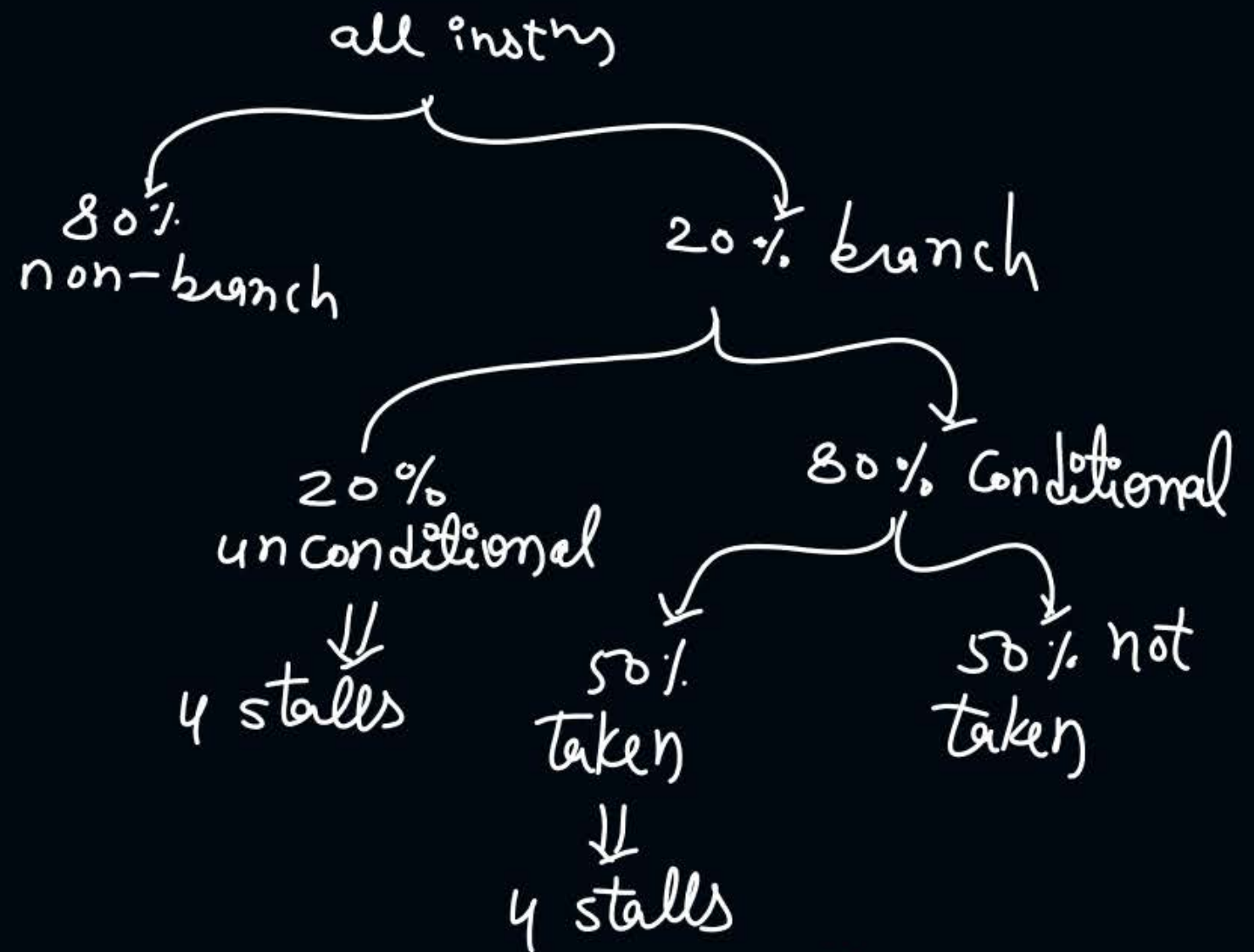
$$1.8 * 2ns$$

$$= 3.6ns$$

$$CPI_{avg} = 1 + 0.2 * 4 = 1.8$$



2.



$$CPI = 1 + (0.2 * 0.2 * 4) + (0.2 * 0.8 * 0.5 * 4)$$

$$= 1.48$$

$$A.I.E.T. = 1.48 * 2ns = 2.96ns$$

GATE - 1st YQ

#Q. A processor X_1 operating at 2 GHz has a standard 5-stage RISC instruction pipeline having a base CPI (cycles per instruction) of one without any pipeline hazards. For a given program P that has 30% branch instructions, control hazards incur 2 cycles stall for every branch. A new version of the processor X_2 operating at same clock frequency has an additional branch predictor unit (BPU) that completely eliminates stalls for correctly predicted branches. There is neither any savings nor any additional stalls for wrong predictions. There are no structural hazards and data hazards for X_1 and X_2 . If the BPU has a prediction accuracy of 80%, the speed up (rounded off to two decimal places) obtained by X_2 over X_1 in executing P is

1.43 Ans.

$$S = \frac{CPI_1 * \cancel{tp_1}}{CPI_2 * \cancel{tp_2}}$$

$$= \frac{1.6}{1.12}$$

$$= 1.43$$

$$CPI_{X_1} = 1 + 0.3 * 2$$

$$= 1.6$$

30% branch

20% wrong
 ↓
 2 stalls

$$CPI_{X_2} = 1 + 0.3 * 0.2 * 2$$

$$= 1.12$$



Topic : Instruction Pipeline

- IF: Instruction Fetch
- ID: Instruction Decode & Address Calculation
- OF: Operand Fetch
- EX: Execution
- WB: Write Back





Topic : Classic RISC Pipeline



1. IF — Instⁿ fetch
2. ID — Instⁿ decode
3. EX — Execution
4. MEM — memory access
5. WB — write back





Topic : Classic RISC Pipeline for Computation → ADD, SUB, etc.

1. IF — Instⁿ fetch from mem. & PC increment

2. ID — Instⁿ decode and register read (operand fetch)

3. EX — ALU operation

stalls due to ^{each} data dependency = $5 - 2$
= 3

4. MEM — Nothing

5. WB — Register write (write back result to destination register)



Topic : Classic RISC Pipeline for Load

Reg. ← Memory

1. IF — Instⁿ fetch & PC increment
2. ID — Instⁿ decode
3. EX — *Nothing*
4. MEM — Memory Read
5. WB — Reg. write



Topic : Classic RISC Pipeline for Store

memory ← Reg.

1. IF — Instⁿ fetch & PC increment
2. ID — Instⁿ decode & Reg. Read
3. EX — Nothing
4. MEM — memory write
5. WB — Nothing



Topic : Classic RISC Pipeline for Branch

1. IF — Instⁿ fetch & PC increment
2. ID — Instⁿ decode | E.A. calculation | Condition check | PC updation if needed

3. EX — Nothing

4. MEM — Nothing

5. WB — Nothing

No. of stall cycles for each branch instⁿ

$$= 2 - 1$$

$$= 1$$

#Q. Consider a program which contains 50 instructions I1, I2 , I3 I50. Further consider a 5-stage classical RISC pipeline with stages as: IF, ID, EX, MEM and WB. The program contains only 1 branch instruction which is instruction I5, and its target is instruction I48. If during the execution of the program the branch is taken, then number of cycles required to execute this program in the given pipeline is 13 ?

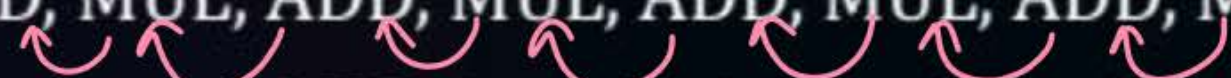
$$n = 5 + 3 = 8$$

$$\begin{aligned} \text{w/o stalls} &= 5 + 8 - 1 = 12 \\ \text{stalls for branch} &= 2 - 1 = 1 \\ \hline \text{Total} &= 13 \end{aligned}$$

- #Q. Consider a pipelined processor with 5 stages, Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Write Back (WB). Each stage of the pipeline, except the EX-stage, takes one cycle. Assume that the ID stage merely decodes the instruction and the register read is performed in the EX-stage. The EX-stage takes one cycle for ADD instruction and two cycles for MUL instruction. Ignore pipeline register latencies.

Consider the following sequence of 8 instructions:

ADD, MUL, ADD, MUL, ADD, MUL, ADD, MUL



Assume that every MUL instruction is data-dependent on the ADD instruction just before it and every ADD instruction (except the first ADD) is data-dependent on the MUL instruction just before it. The Speedup is defined as follows:

$$\text{Speed up} = \frac{\text{Execution time without operand forwarding}}{\text{Execution time with operand forwarding}} = \frac{30}{16} = 1.88$$

The Speedup achieved in executing the given instruction sequence on the pipelined processor (rounded to 2 decimal places) is 1.88.

IF — 1		
ID — 1		
EX — 1	ADD	MUL
	1	2
MEM — 1	stall	1
WB — 1		

with operand forwarding:-

$$\begin{aligned}
 \text{w/o hazard} &= 5+8-1=12 \\
 \text{stalls for structural hazard} &= 4*1=4 \\
 \hline
 \text{Total} &= 16
 \end{aligned}$$

w/o operand forwarding:-

$$\begin{aligned}
 \text{w/o hazard} &= 5+8-1=12 \\
 \text{stalls for structural hazard} &= 4*1=4 \\
 \text{--- 11 --- data hazard} &= 7*(5-3)=14 \\
 \hline
 \text{Total} &= 30
 \end{aligned}$$



2 mins Summary



Topic

CPI Calculation

Topic

Classical RISC Pipeline





Happy Learning

THANK - YOU

